

Star Dunes Research Meeting - Meeting Recording

July 21, 2026, 1:02PM

42m 52s

● **Author** started transcription

Author0:03

It has started recording and transcription.

OK, so back to the presentation.

Yes, can you see my screen?

Participant 2 0:18

Yeah.

Participant 1 0:19

Yes.

Author0:19

Yeah.

So for the feedback, first of all, I want to get a basic background of like, what is your expertise or what you consider is the thing that you have worked on the most and what other natural environments other than the desert environment, if you have ever like researched.

So.

You can go ahead just one by one or yeah.

But this is like a discussion session, so anyone can start with.

Just what you would say is your expertise.

Participant 2 1:00

Well, my main focus is on geomorphology and in desert environments. So that's probably my expertise. Yeah, other natural environments are probably, yeah, mostly deserts.

Author1:06

Okay.

Participant 1 1:13

Okay.

Author1:17

Yes.

Participant 2 1:18

Yeah, but in our working group, we look at a few different environments. One group works mainly on coastal geomorphology, with regards to the question of paleo-environmental changes, mainly bioclimatological changes. Another group recently started looking at fluvial and glacial.

Author1:50

Yeah.

Participant 2 1:59

changes in the Delta.

looking for the slumping sites there where peri-glacier permafrost is now thawing.

And that region was a lot of, let us say, morphodynamics because of climate change.

But the main area what we are looking for is

drylands. So most of us are working in drylands one time and we have one project more or less with fluvial geomorphology, but in historical times. So the question, what was human impact on the fluvial system?

Author2:42

Okay, yes. Thank you for your input.

Okay, then I would assume mostly it's geomorphology and geodesy.

Participant 2 2:54

Yeah, and a little bit, a little bit sedimentology.

Author2:54

Okay.

Participant 1 2:55

Mm.

Author2:57

And sedimentological.

Participant 2 2:57

So, especially what what model is also doing is because of the sense in dry lands, so mainly the grain size distribution, the colour, and also the maybe the

Author3:13

Okay.

Play.

Participant 2 3:18

Mineral composition of sense.

Author3:22

Okay, yeah, that should be informative. Next is about the temporal encoding. If the colour progression was clear, if you could tell before looking at the legend, like what is like, which is the oldest and which is the earliest, like

What do you think about the colours used for showing the layers?

Yeah.

Yeah, that's why you're unmuted. Yes.

Participant 3 3:55

Yeah.

I would change the crest movement between red and green.

Participant 1 4:00

It.

Participant 3 4:03

With regard to colour blindness of people with red and green.

Author4:04

Okay.

Yes.

Participant 3 4:09

Disease.

Author4:11

Yeah, to red and blue. OK.

Participant 3 4:12

I mean, it's quite...

And also from a geomorphological perspective, thinking about elevation models, we always suggest that red colours are higher than green colors. In this case, red would be positive and green would be negative if you stay with red and green.

Author4:25

Okay.

Okay, yes, I got it.

Participant 3 4:34

But this is only my, I mean red means negative, green is positive, but with regard to transition it in elevation data, so, and then I would suggest to change it from red to not to use red and green.

Author4:37

Yeah.

Yes.

Participant 3 4:51

Maybe. Participant 1 is...

Author4:52

Yeah.

Participant 1 4:52

But.

No, I agree with you, but even what we're looking at here is a horizontal movement.

So a plus or a positive and negative is not so intuitive, I think. So maybe...

Author5:07

You.

Participant 1 5:08

Colours are maybe good, but maybe you have to use completely different ones that we do not associate with the position erosion or topography itself, because it's a lateral movement.

Author5:15

Positive.

Okay, understood.

Participant 2 5:26

I would say to scale to scale to use a scale of change. If you if you put colours on it, then so higher changes will be a more bright color, maybe something with regard to, I don't know.

Author5:26

But...

Participant 2 5:47

Violet also, something like that, and low changes, maybe not so bright colours or yellow or so, I don't know.

Author5:58

Yeah.

Participant 2 5:59

So that you can see not only the arrow lengths with the changes, but also the colors.

Author6:09

Yes, OK, yeah.

Okay, so...

Participant 2 6:13

or just use one colour with different intensity.

Also possibility.

Author6:17

Yes.

Ah, for at least the movement, I always thought because it moves like either ahead or behind like advanced retreat, positive, negative, which is why a more diverging way is preferable. But yes, I will take that into account as well when I'm thinking about the next.

Participant 2 6:36

Yeah, but positive, negative is depending on which direction. So you have a lot, it could possibly in every direction. So in which direction is it positive, in which direction is it negative? And that's depending on the crest line you're at where you're plotting the movement. So that's maybe a little bit difficult to implement.

Author6:51

You.

Okay, okay, yeah.

Participant 2 6:59

Plus minus only for high changes.

Author6:59

I think this.

Participant 2 7:02

So, altitude and challenges, because the crest line is not moving negative, it's just moving in a different direction, so that that.

Participant 1 7:03

Mhm.

Author7:03

Ahh.

Okay, that is.

Yes, more right and left.

OK, so.

Participant 2 7:19

Yeah, I know.

Participant 3 7:22

But, so, basically, the thickness of the arrow and the colour scheme.

Author7:22

Okay.

Participant 3 7:28

is based on the same attribute, like the metres and change.

Author7:33

Mm.

Yes.

Yes.

OK, so.

Then I'll move to the next, which is about the visual hierarchy. If you could just tell me what was the first thing that stood out after you went to the dashboard.

Participant 2 7:58

suppliers. That's what I first look at because it's the biggest feature in colour probably.

Author8:00

Hey.

Participant 3 8:02

Yeah.

Author8:04

Yes.

Yes.

Participant 2 8:07

No, but it just depends.

Author8:08

FBI put in.

Participant 2 8:10

Yeah, it was my first time.

Author8:10

Yeah, yeah, but...

Participant 2 8:12

That was the question.

Author8:13

Yeah, but every input has, like, because everyone will have a different thing or...

Is what I would expect, but yeah, but sort of be first, like...

Yeah, all inputs from everyone would be helpful.

Participant 2 8:31

Let's do it as the same, but so is that as Office Final dashboard?

Participant 3 8:32

Yeah.

Participant 2 8:37

You are, are you? Yeah, that's correct.

My first impression was the crests.

Author8:47

OK.

Participant 2 8:47

That one, but, but yeah, this is because of experience, so we looking for dunes, and this is why I'm always looking, first of all, for linear structures for the crest lines. I learned it in that direction.

Author9:01

Yes.

Yeah, it does.

Participant 3 9:06

I don't know if it was my first or my second view is that it's quite obvious that the playa polygons are not the same extent as the white playa areas.

I think it's probably the it was probably the second view, but I find this quite obvious that it doesn't match too much for the aerial extent.

Author9:33

Yes, okay. Yeah, that is mostly because I have only used the 9th cent percentile, but yeah, that helps. Okay.

Participant 3 9:40

Yeah.

Participant 2 9:40

So, we're still not.

Author 9:44

OK, so then I.

Participant 2 9:45

Six.

Participant 1 9:46

But this...

Like this regard, I had a question disregarding the accuracy now, is the playa information useful?

Participant 2 9:48

which was excellent, it's fair to call up.

That's a problem about which you can't.

Participant 1 9:55

The mapping, let's say.

Participant 2 9:56

But doesn't it?

One again.

Participant 3 10:02

I would.

Participant 1 10:02

Cure, I'm sorry.

Participant 3 10:04

I wouldn't say so because if you have these temporary changes with the different polygons,

But...

Author 10:11

Mhm.

Participant 3 10:13

So the actual impression about the playa is that these index is not covering the whole playa area.

Participant 2 10:17

Let's all.

Participant 3 10:19

Oh.

Participant 2 10:21

Yeah.

Participant 3 10:21

It might be difficult to compare different times of the year if the impression that is not actually mapping the whole extent of the playa.

I think it's probably mapping this index a very specific part of the playa, which can be of interest also in the time interval.

But therefore probably you have to understand the index, what is actually mapped.

Participant 2 10:43

What's next?

Participant 3 10:49

Because I don't think it's a whole playa, it's a different colour or different spectral information, which is not the same as the playa, at least in our geomorphological perspective.

Author 10:50

Okay.

Yes.

Yeah, yes.

Okay.

Yeah, then moving on to the next one is, were the preset modes useful and intuitive to use, and are there any?

The other way to you can say, like, it's better to like in temporal periods.

So, if yeah, or the piece I was that were the compare annually, monthly, and custom.

Yeah.

So, if you...

Participant 2 11:42

I did not see the preset modes, but I was, I was looking at Participant 2's screen.

Yeah, where the presets mode be useful and intuitive for you. That one's here, prior or the preset modes. I don't know exactly what it is, so...

Author 11:47

I see.

Participant 3 11:55

Hey.

Author 11:56

Okay.

Okay, okay, bye.

Participant 3 12:03

Actually, I cannot change date B.

Author 12:06

Oh yes, yeah, that is I have set it to only consecutive dates at the.

Yeah, the dashboard only has consecutivities because the wind data, I felt it changes and it oscillates from back and forth. So for compare preset, yes, you cannot change the date B, but that's why I had the presets like you can otherwise choose other months, years, customise it however you want, but.

Then I will add the customization also during compare for both days.

Participant 3 12:40

Yeah, because that would be the most intuitive one to say, okay, I have to start date one and a second date to compare like 2017 to now or to 2020. This would be the most intuitive tool I would use to compare two dates, which I choose by myself.

Author12:58

Okay, I understand. So it is better than to have the consecutive reads. Okay, yes.

Participant 2 13:02

Mm.

Author13:04

But...

So the preset mode was not in the, like, or the different, like, you can do it annually, see the quest annually or monthly was during the exploration, not really seen.

Participant 3 13:19

No.

Participant 2 13:19

Yeah, maybe you can add a seasonal preset, but...

I think I will monthly is also good.

Participant 3 13:27

Because so far I found no solution to see the difference between the first image and the last image, between 2017 and 2026.

Author13:36

Ah, okay. Yes, I see. Yeah.

Participant 1 13:36

Mhm.

Author13:42

Okay, yeah, that is not possible.

Thank you for pointing that out. Previously I had the customization, but in the process of then updating onwards with the more feedback I got, yeah, I may have missed on that point. But yeah, I will include that change as well.

Participant 1 13:45

I.

Author14:06

It.

Participant 1 14:07

If it's possible, I would even prefer a slider, a time slider rather than a drop-down with the dates. But I think it's a cosmetic thing, maybe.

Participant 3 14:18

Yeah.

Author14:20

Okay, yes, yeah, I would try that as well. But what I noticed was there are a lot of dates and the space for the slider was not enough at that in that column. So if you want a specific month and a specific year, you have to really go over the slider like

Participant 1 14:33

Ohh.

Author14:41

very minute movement with the mouse.

Participant 1 14:42

Okay.

Author14:46

Which is why it is a drop-down.

But, OK, yeah, that informs me about my.

Participant 3 14:53

Can you all, instead of a slider, can you also use a calendar instead?
Of the drop-down.

Author15:03

I will have to look into it. I have only been using the toggles that is that are provided by the streamlit interface, so I am kind of bound by what is possible.

Participant 3 15:08

But...

Participant 2 15:12

The.

Participant 3 15:17

OK.

Can you?

Yeah.

It's OK, but in the timeline, you have a lot of more dates to scroll down to go to the end, maybe just, but...

Author15:32

Yes.

Participant 1 15:34

But you can type in the filters, and that's good.

Author15:35

Yeah.

Yes, in the drop-down box, you can actually type the date.

Maybe that needs to be iterative.

H.

Participant 3 15:46

If you know the date of acquisition.

Participant 1 15:50

No, but if you type 2025, it shows only the 2025. If you type a month, it shows only one option.

Participant 3 15:50

Let me.

Oh yeah, yeah, yeah.

Author 15:57

Yes, you, yeah, don't need to know till the date, yeah. Okay, that was quite informative. And then to the next question, it is about the errors and gaps visualization, did seeing the margin of error like directly on the map help you with like understanding that this detect.

Participant 2 16:04

Yes.

Author 16:18

Crest lines are not the accurate version, so they did add value for you. is the main question and were the data gaps clear from the start.

Participant 2 16:35

You're on the phone.

Participant 3 16:42

You mean that it's obvious that some of the crests are not digitized?

Author 16:48

Yes, or yes.

Participant 2 16:52

Yes, it is obvious.

Participant 3 16:53

Look.

It's clear.

Participant 2 16:55

And the wind data is also obvious, the coverage of the wind data.

So, that is not completely continuous.

Author 16:59

Okay.

Participant 2 17:04

So for me, it's clear the question is, so we are, so we are educated and that, well, the question do you ask for, let us say, not so much educated people, then maybe

Author 17:05

Yes.

Participant 2 17:21

So sometimes I'm showing my parents also or my sister or my wife maps also. And for us, everything is clear, but for them, something, sometimes nothing.

So this is a question of what knowledge is there.

So for me, everything was clear.

Author 17:42

Yes, yeah, that does play the main factor in like people usually assume that the maps are completely right and the trust in maps. So to give the more better accuracy of it, like there are some gaps or that was my main idea behind.

adding the errors and gaps into the visualization. So were they clear enough was yeah.

Participant 2 18:06

It's 11.

Author18:09

Depends on the users, yeah.

Participant 1 18:17

I think you meant less that there are gaps where there is nothing but that there are these dashed lines, right?

Author18:23

Yes, but I suppose you also have to turn on the toggle for that to see the continuity of the line.

Participant 1 18:29

Mm.

Author18:31

But yeah, it depends on how you explore the dashboard. If you did not see the need to even turn on that like dotted lines, the gap fills, then that also informs me about like if they are necessary or not.

Participant 1 18:49

But maybe they should be toggled by default because people...

Author18:54

Okay.

Participant 1 18:54

Seem to miss it. It's still loading for me, but...

Author18:59

Okay, yeah.

The.

So that was my main point for the error and gaps visualization. If anyone has anything to add on to the margin of error.

as the buffer for requests.

But that also, yes, it is not on by default. It is all turned off in the hopes of not bombarding the user with too much information all at once.

But.

If there are no additions to this point, then I will move ahead to the next question.

which is about the wind rose graphics. Were they helpful? Or did it like take up space on the map without providing any useful information because the data is from like a farther away wind station. What were your thoughts on this?

Yes, Ben. Yeah.

I think so, you're muted.

No.

Mm.

Participant 3 20:30

For me, it wasn't clear that the wind rose data.

Author20:31

Yes.

Participant 3 20:37

It's actually the same time frame than the satellite image.

Author20:44

Okay, so yes.

Participant 3 20:45

It wasn't obvious for me that the Wind rose data also change if I change the dates.

Participant 2 20:52

Just checked it.

Author20:56

Okay.

Yes, the wind rose data was based on the layers shown on the map. So whichever 2 dates you have selected, that's what it uses to make the wind rose.

Participant 3 21:03

Mm.

Participant 2 21:10

Maybe you should add.

Stay to the wind rose, it would be better to have a station nearby, but it's not existent, so, but it's also expert knowledge that you know, okay, the weather station needs to be directly there, not 100 kilometres away. It's not visible in the in the display, probably, as long as you don't have an overview map where the weather station is.

Author21:14

Okay.

Okay.

Participant 2 21:34

That kind of marked on the on the overview map or something like that.

Participant 1 21:39

But I think it's an important point. So maybe, you know, sometimes in maps you have the thing and then you move to a different location and it's still there, but it kind of points to where it comes from. Because we have to think in the future, we have maybe 3 stations at once. So we need the location mapping where each of the wind roses comes from and then maybe one is from.

Author21:40

Okay.

Yes.

Participant 1 22:02

90 kilometres away and it needs to be somehow indicated. So I think that would be cool.

It doesn't have to be so dynamic, but to at least be able to see, this is the wind rose is not located in the map position on the upper right. It's just there to see it.

Author22:08

Are you?

Okay, yes, I will look into if it is possible with the folio map. But also in this dashboard, I had included the wind station data, weather station data from 16th of March to 20th of March. And I have like directly added the location for the wind rose for that.

Participant 2 22:32

Yeah.

Participant 1 22:33

Ohh, no, I see it.

Author22:41

But okay, so for Wind rose, I will add the dates for which dates were used to create that.

Wind rose. Then I'll move on to the next question, which was like, oh, yes.

Participant 3 22:51

And so, sorry, back again to the wind rose. So, if there is a selection of several months, there are several wind roses.

Participant 1 22:52

Hut.

Author22:55

Yeah.

Yes.

Participant 3 23:02

And then it's moved from the map to the right part and you have several wind roses. Why is it?

separated into different ones.

and not the accumulative wind rose for these four months or whatever.

Author23:22

Yes, so at the, in the different monthly and annually phase, the wind roses are moved to the side and only shown between 2 consecutive dates is the, my thought process behind it was that you can see all the crest lines at once for four months, let's say.

And the crest move ahead or

Participant 2 23:24

Seven fits vector.

Author23:43

let's left or right and I wanted them to be separate so you can see from one crest to the other what was the direction and speed.

Participant 3 23:52

Yeah?

Author23:54

And from the next crest to the consecutively next crest, what was the wind direction and speed? I felt that that would be more informative than a cumulative wind rose where you can't say like, yeah, so that is why it's separate. Does that answer your question?

Participant 2 23:58

I was little, what is wrong?

One of it was even us.

Participant 3 24:14

Yes. And then, then, but if you have 4 crests, I have now 4 months and I have 4 crests in different colors. I only can see which colour is which month if I point on it, because with this colour scheme, it's from little to orange and the blue one in the middle.

Author24:20

Mm.

Yes.

Yes.

Participant 3 24:35

I have to point on it.

Participant 2 24:36

Yeah.

Author24:36

Mhm.

Participant 3 24:38

Can you change the legend that you have a distinct colour table for each line?

Author24:47

Okay. Yes, I can. Yes, I understand what you mean because the middle section is a bit like if it is going ahead or not. I can make it only a singular colour from like, like from white to a concentrated color. So the brighter it is.

Participant 3 24:48

You know what I mean?

Yeah.

Author25:07

the more accurate to the colour it is than it is the latest. I can change the colors, yes.

Participant 3 25:14

I mean, the colours are for me okay, but if I have maybe 10 lines, that I have also 10 entries in the legend which is distinct colour and not only the colour scheme.

Author25:20

Yes.

Okay, yes, that would be a challenge to add it like that, like separate colour for the

separate dates. Yes, because it keeps you, this is a dynamic dashboard, so the dates will also keep changing. Then I can look into it, but then I will have to.

Participant 3 25:37

For each, yeah.

Author25:48

to stop the amount of layers seen or amount of time seen to like only four or six epochs because I will have to accordingly update also the legend. But I can look into it, but for now, having a sequential like ramp, colour ramp was the best solution.

Participant 3 25:55

Yeah, and...

Yeah.

Author26:12

That.

So, yeah, if anyone has anything more to add on to this, or...

And.

I will go to the next, which was about design choices. Just overall, how you feel looking at the dashboard. Does it scream like, oh, this is about the deserts, the dunes. It's a general design, so it is subjective opinion question.

About if the planet.

Participant 3 26:41

Yes.

Participant 2 26:43

Yes, it's fine.

Participant 3 26:44

Ha.

Author26:45

Did.

Participant 1 26:45

No, no, no, I don't accept this answer because I would like to see it in the future or when Author completes really as a web viewer that I would like to host and link to our project. So now imagine it, there's the logo of related project universities maybe.

Participant 3 26:48

Yeah.

Author26:49

Big.

Yes.

Participant 1 27:06

What would you like to change in the design? So this is maybe the question here.

Author27:13

OK, yeah, the colour choices should it be like?

Participant 3 27:14

And so you have a blue, you have a blue label and a red one.

Participant 1 27:22

I don't mean for the logo, I just mean it's not only how it looks nice for this work, but in the big scheme, what do you think about the design?

Participant 2 27:22

Yeah.

Author27:22

Yeah, and...

Participant 3 27:24

Yeah.

Yeah.

Author27:31

Yes.

to use the colours of the logos, to use it as the accentuated colors. So for the slider or for the drop down or the radio buttons, you change it to the blue or red. And yeah, that is the design choices.

Participant 2 27:47

No, generally, if it's if it's directly linked to projects and the university site is brought, it should probably be more neutral from the colour like or whitish in the background.

Author27:57

Okay.

Participant 2 28:00

Well, all white in the background, maybe.

Author28:00

Yeah.

Yes.

Participant 1 28:06

I mean, it will not be embedded. It would probably still be somehow be an external thing. Yeah. So to me, I would also use a more...

Participant 2 28:08

Yeah, but I know what you mean.

Participant 1 28:17

standard font, for example. So it looks very nice. It has this map style, but I think for a live dashboard, maybe something like what you use in the presentation here or this standard Helvetica or whatever would look, yes.

Participant 2 28:30

Yeah.

Participant 3 28:32

Without the reason.

Author28:34

Yeah.

Participant 1 28:35

Less, less Schnakelig.

Participant 3 28:37

Yeah.

Author28:38

Yes.

Participant 2 28:39

Yeah, more neutral. Yeah, I know.

Author28:41

Yes.

Okay, I will. Yeah. Okay, so final is the overall assessment, do you think it effectively communicates the years from 2017 to 2026, how it changed? It's more of like, yeah, yes or no, if it helps according to you.

Participant 3 29:06

So actually I find it really, really nice to play around with it, to set different dates and then to see the press lines, etc, etc. I would skip the other in-situ layers, which makes it a little bit more complicated between, because this is just more or less random information, which are not necessary, helpful for the dual trusts.

Author29:31

Okay.

Participant 2 29:36

Yeah, and I think it affects, yeah.

Author29:36

Any other info?

Participant 2 29:39

I think it communicates good, but you have to know what you're looking at and what you're looking for, most likely. You have to know about units, you have to know about press movement, and then you can interpret the map. But that's probably, yeah.

Author29:55

Okay, yes, yeah.

Participant 2 29:56

And story kind of.

Author29:59

Yes, my users were always the project members, which is also the way I would have thought about how to communicate, yeah.

Participant 3 30:11

And I think it's in the end, it's so specific that also everyone which will be using it is also not a family member or it kind of expert which is entering such a website and playing around with it. So I think you can have these

Author30:12

OK.

Participant 2 30:28

I have it.

Participant 3 30:30

Pre-knowledge about the people who which are using it.

Author30:36

OK, yeah.

Participant 3 30:37

But probably because it's also about starting dynamics, you really think should think about the playa mapping because it's also not in the title, it's not in the general with the wind roses and blah blah blah. So why is it mapped and which information is it actually added value in this dashboard, which is about the crest movement.

Participant 1 30:37

And we can use it.

Participant 2 31:02

True.

Author31:02

Yes.

Participant 1 31:08

Just to add for your transcript, the experts can use this to make a map which can be exported. I don't know if this was noticeable. And then you can show it to the audience that you have with whatever detail you need or not.

Participant 3 31:16

Mm.

Author31:24

Mm.

Yes.

Participant 3 31:27

But can you also export this as an image file, because now it's only HTML, so you can play around with it again with this view, but not to copy it as an image file also.

Author31:41

Yes, currently it is not possible to download it as an image. Again, that comes with the Python libraries. Yeah, I will have to look into it, but you can still export it as HTML and embed it into your projects. Like if just to show, just to inform the audience.

for specific dates and so on. But it only exports the map layer. Yeah.

Participant 3 32:06

Yeah, because I think most of the users, if there's a download map function, I would always expect an image file.

Participant 2 32:09

One is good.

Participant 3 32:15

Author32:15

Okay.

Participant 3 32:17

Information.

Participant 2 32:18

Thanks.

Participant 1 32:19

Mm.

Author32:19

Yes.

Participant 2 32:20

Yeah, and together with the download, I think you have to also bring in the scale, if possible.

Participant 1 32:25

Yeah.

Participant 2 32:29

So if I downloaded this as HTML, I'm missing the scale.

Author32:35

Okay.

Yes.

Um...

Participant 2 32:43

And then can you move, can you move it? Can you rotate the screen or is it always the top is always north?

Participant 1 32:43

I...

Author32:43

Okay.

Participant 2 32:54

Is it?

Author32:54

Top is always north.

Participant 2 32:57

Okay, I think then in my opinion, you do not need a north arrow, but some colleagues are looking always for such errors. Where is north?

Author 33:03

Yes.

You.

Participant 3 33:08

Yes, of course you need it because there is no coordinate grid.

Participant 2 33:10

No.

Participant 1 33:11

Mhm.

Participant 2 33:13

Yes, so, but you can imagine if this north is always up, then OK, this is a regular rule, however.

Participant 3 33:21

But we are on the Southern Hemisphere.

Participant 2 33:22

Yeah, yeah, of course. And all maps are looking to the south and all the or to the east, especially.

Participant 3 33:25

Yeah.

Participant 1 33:30

You can just read the wind rose and you know.

Participant 3 33:33

Yeah.

Participant 2 33:33

Mmh.

Author33:33

Yeah.

Participant 3 33:34

But why are there two maps in the HTML?

Participant 2 33:39

Do we have? We do not have.

Author33:40

Um...

Okay, I will have.

Participant 3 33:43

So if I scroll down on the right side, there's another map below.

Author33:49

Okay, that was.

Participant 1 33:50

The large base map, yeah.

Author33:53

Yes.

Participant 3 33:53

But also including the polygons, it's duplicated.

Participant 2 33:54

Is on the screen again?

Author34:00

Oh, I will have to look into it. Thank you for pointing that out. Yeah. Okay.

Participant 2 34:07

No.

I want to say it's gone.

Author34:09

But.

I'm sorry, that is an error left behind by me. I will look into how the HTML is being exported.

Participant 3 34:14

No problem.

Yeah.

Author34:20

Thank you for pointing that out. Okay, so my last question was about other suggestion, but I think so I also got a lot of suggestions about how to improve it, how to change it, or yeah. So if there is anything like to add,

Participant 2 34:23

System.

Participant 1 34:38

Yeah.

Author34:43

Or, I can.

Participant 2 34:45

Yes, but you can.

Participant 3 34:47

I mean, what is...

Participant 2 34:48

I mean.

Author34:49

Mmh.

Participant 3 34:50

For me, looking very well is actually the detection of the crests. I mean, this is now a side work more or less, but because it's not all always about the visualization, but

Author35:02

Yes.

Participant 3 35:03

The most interesting stuff for me is actually these changes based on these on the pixel detection.

Author35:13

Yes.

Participant 3 35:13

But I don't know if this is the main focus of your work, probably not.

But I've...

It would be for me the most interesting work, which is looking so promising.

Author35:28

Yes, it is always important, yeah.

Okay.

Participant 1 35:34

It comes with a package, yeah.

Participant 2 35:34

Yeah, and I will.

Author35:34

Just been.

Participant 4 35:35

Uhh...

Participant 2 35:36

I would add also from a geomorphological perspective, the wind data, maybe you can add another wind rose or even a sand rose only displaying the wind which can potentially move sand. So maybe wind always above 5 metre per second.

Or, if you can visualise this in the in the uh rows, or even, yeah, sand rows is even more difficult if you would like, uh, just calculate the potential to uh.

Uh, relocate sand to move sand, yeah, but I think...

A step in the middle would be to reduce it to the winds capable of loading sand, so all above 5 metres per second.

of 4.5 or depending on what threshold you set.

Author36:23

Okay.

Yes, okay, yeah. Yeah, thank you for all your feedback. I will look into what I can now apply into my dashboard in the limited time I have. And then in a short time, I will share the dashboard again with a separate Google evaluation form.

So I would be really grateful for as many inputs as you can. Yeah, provide the feedback as possible.

So, that's all from my side, but yes.

Participant 4 36:54

And just...

One question, maybe from a more amateur side: Is it possible to also have the viewed area or the area of interest in color, or why is it black in or in grayscale?

Author37:14

That is the base imagery, satellite base imagery. But if you don't want to see the black and white base image, the satellite image, in the toggles in the remote sensing layers, at the very bottom, you have a base image which you can toggle off. And then it will just show you the

Participant 4 37:29

Mhm.

Author37:33

Yeah, the satellite-based map that I have from SV.

Participant 4 37:34

Okay.

Okay, no, I said. Thank you.

Participant 3 37:38

But that's a good point, but there is probably a lot of data. Would it be possible to actually display the used Sentinel image?

Where the crest mapping is based on.

Author37:52

The Sentinel image, the black and white Sentinel image is exactly the near infrared image that was used for the crest edge detection.

So if you also see like which year the image is from, which is shown at the bottom in the legend.

Participant 3 38:09

Base image.

Author38:13

You can see the...

Participant 3 38:14

Yeah, but now I have a philtre 2017 April to August and the base imagery is from January.

Author 38:22

Yeah, that is a yeah, a bug for my annual.
preset. I will, yeah, correct it shortly, but for if you are in the compare preset or if you just change around the months, it should reload to have the exact, the oldest crest lines imagery beneath it.

Participant 3 38:28

Okay.

Participant 1 38:41

Mmh.

Participant 3 38:50

Oh yes, now it's changed with the compare setting, not in the enabled.

Participant 2 38:56

Ohh, it's working.

Author 38:56

Yes.

So, the, yeah, the data, the satellite image used, everything is all.

Participant 2 39:10

So they say, hold on, let's in the one cookie.

Participant 3 39:17

And maybe, so I just see it now. So if you have the crest movement and the colour scheme and the thickness of the arrow, could you include for colour the angel of movement? If you have a colour scheme from zero to 360 degrees, then you have actually a direction

Author39:36

Yes.

Participant 3 39:39

For the colour and the thickness for the amount.

Author39:44

Okay.

Okay, for from orientation.

Participant 3 39:48

So this would be an additional information because there's always a direction. If you point on it, the orientation is there. But if you visualise the orientation instead of having two visualisations for the crest movement.

Participant 2 39:53

Send me a nice thing.

Author39:56

Yes.

Participant 2 39:57

So, if I understand.

Author40:05

Okay.

Yes, that is.

Yes, that would be interesting to try.

Participant 3 40:13

And then you don't have the issue with what is positive, what is negative, in which direction, north to south, and back and forth of the crust.

Author40:14

Yes.

Participant 2 40:19

Set up our meeting, so we see.

Author40:22

Uh, yes, that that would be an, yeah, yeah.

Participant 2 40:23

I come on to hold out that I mean, it is so super.

Participant 1 40:27

It also adds a bit visual complexity, because you also see the direction from the arrow, obviously, but maybe try it out.

Participant 3 40:33

Yeah.

Author40:37

Yes, if it's.

Participant 1 40:38

Like a circle legend.

Participant 2 40:39

Okay, okay.

Author40:42

Yeah, or maybe it can also be based on the like, not just the 8th direction.

Participant 2 40:46

Suddenly, you know, listening.

This is a pixel.

Author40:51

Yeah.

Huh, yeah, that, that.

Participant 3 40:54

Yeah.

Author40:55

Better than the positive, negative, yeah.

Participant 2 40:58

Sing me.

Author41:00

Okay.

Participant 1 41:00

Okay.

Author41:02

Yes.

Participant 1 41:02

Now everyone is playing around with it, so that's a good sign.

Participant 3 41:04

Yeah.

Participant 2 41:04

No, of course, of course.

Author41:06

Yes.

Participant 2 41:07

Very nice.

Author41:08

Okay, okay, then I'll ask.

Participant 1 41:10

Yeah, we already took much more time than intended, so...

Author41:13

Yes.

Participant 2 41:13

Mm-hmm.

Participant 1 41:15

I think continue as long as you want, but thank you already very much for all the inputs. It was really nice to exchange and I think we can repeat it in some time when we have much more progress on the things. Yeah.

Participant 3 41:19

Yeah.

Author41:21

Yes.

● **Author** stopped transcription